

MDM Project on

IoT Based Smart Motor Health Monitoring System

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of the degree of*

Bachelor of Technology

By

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**DEPARTMENT OF ELECTRONICS & COMPUTER SCIENCE
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ABSTRACT

The proposed project, "IoT Based Smart Motor Health Monitoring System," is designed to ensure the reliable and efficient operation of DC motors by providing continuous real-time monitoring of critical parameters such as temperature, vibration, and operating voltage. Industrial and embedded motors are susceptible to degradation due to overheating, excessive vibration, and voltage fluctuations. Traditional maintenance methods rely on periodic manual inspection, which often fails to detect faults in a timely manner, leading to unexpected breakdowns and costly repairs.

To address these limitations, the proposed system integrates Internet of Things (IoT) technology with sensor-based automation. The system employs an ESP32 microcontroller as the central processing unit, mounted directly on or near the motor. Key sensors include the DHT11 for temperature and humidity measurement, the MPU6050 accelerometer/gyroscope for vibration detection, and a voltage sensor module connected through the L298N motor driver to monitor the operating voltage of the motor. All sensor readings are processed by the ESP32 and compared against predefined threshold values. When any parameter exceeds or drops below its safe threshold for example, if temperature rises abnormally, vibration exceeds safe limits, or supply voltage falls too low the system raises an alert. The readings are displayed locally on a JHD204A 20x4 LCD display and simultaneously transmitted to a Firebase Realtime Database. A custom web-based dashboard at <https://smart-motor-health-system.web.app/> visualizes the live data, trends, and fault events, enabling remote monitoring from anywhere.

The system is cost-effective, compact, and scalable. It reduces manual maintenance effort, enables predictive maintenance, and helps prevent motor failure. Overall, this project demonstrates how IoT and automation can be effectively used in industrial motor health monitoring, embedded systems maintenance, and smart factory environments.

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Chapter 1: Overview

1.1 Introduction

The Internet of Things (IoT) has revolutionized the way industrial equipment is monitored and maintained. One of the most critical components in manufacturing and automation systems is the electric motor. Motors are used extensively across industries for driving pumps, fans, conveyors, and machinery. Any unexpected failure of a motor can cause significant production downtime and economic losses. Therefore, continuous monitoring of motor health is essential for ensuring operational reliability and enabling predictive maintenance.

Motor health can be assessed by continuously monitoring key parameters such as temperature, vibration, and operating voltage. An increase in temperature indicates insulation degradation or overloading. Excessive vibration points to mechanical imbalances, bearing failures, or misalignment. Voltage fluctuations suggest supply issues that can damage the motor windings.

The proposed IoT Based Smart Motor Health Monitoring System uses an ESP32 microcontroller as the main controller, mounted on the motor body. The DHT11 sensor continuously measures the ambient temperature and humidity around the motor. The MPU6050 inertial measurement unit captures real-time vibration data. A voltage sensor, connected through the L298N motor driver, monitors the supply voltage to the motor at all times. All sensor readings are shown on a JHD204A 20x4 LCD display for local viewing and simultaneously transmitted to a Firebase Realtime Database. A web dashboard at <https://smart-motor-health-system.web.app/> provides live monitoring, trend graphs, and instant fault alerts whenever any parameter crosses its predefined safe threshold.

This system eliminates the need for manual inspection, reduces maintenance costs, and provides early warning of potential motor failures, making it highly suitable for smart factory and Industry 4.0 applications.

1.2 Problem Statement

Electric motors are the backbone of many industrial and embedded systems. However, motor failures remain one of the most common causes of unplanned downtime in industrial environments. Most conventional motor maintenance strategies are either time-based (periodic inspection regardless of actual condition) or corrective (repair only after failure occurs). Both approaches are inefficient time-based maintenance wastes resources even when the motor is healthy, while corrective maintenance results in costly and unpredictable shutdowns.

The major challenges with existing motor maintenance practices include the absence of real-time monitoring of key health indicators such as temperature, vibration, and voltage. Overheating is one of the most common causes of motor insulation failure, yet it is rarely monitored continuously. Mechanical vibration due to bearing wear or imbalance progresses gradually and is difficult to detect without continuous sensor data. Voltage irregularities, such as under-voltage or over-voltage conditions, can also silently degrade motor performance and lifespan.

Furthermore, traditional monitoring systems tend to be expensive, complex to deploy, and not accessible remotely. Maintenance personnel must be physically present to check motor condition, which is impractical in large facilities or remote installations.

Therefore, there is a clear need for a low-cost, IoT-enabled motor health monitoring system that can continuously track temperature, vibration, and voltage, compare them against safe thresholds, trigger fault alerts in real time, and make the data accessible remotely through a web dashboard. The proposed system is designed to fulfill all these requirements.

1.3 Objectives

The main objectives of the proposed IoT Based Smart Motor Health Monitoring System are as follows:

1. To design and develop a smart motor health monitoring system using IoT technology and an ESP32 microcontroller.
2. To continuously monitor the temperature and humidity surrounding the motor using the DHT11 sensor.
3. To detect and quantify motor vibration in real time using the MPU6050 accelerometer/gyroscope module.
4. To measure the operating voltage of the motor through the voltage sensor connected via the L298N motor driver.
5. To process all sensor data using the ESP32 and compare values against predefined safe thresholds for fault detection.
6. To display real-time sensor readings and motor status on the JHD204A 20x4 LCD display.
7. To transmit all sensor data to Firebase Realtime Database for cloud-based storage and retrieval.
8. To develop and deploy a web-based monitoring dashboard at <https://smart-motor-health-system.web.app/> for remote access.
9. To generate instant fault alerts on the dashboard whenever temperature, vibration, or voltage crosses safe limits.
10. To create a cost-effective, user-friendly, and scalable system applicable to industrial, research, and educational environments.

Chapter 2: Literature Survey & Proposed Work

2.1 Introduction

A literature survey helps in understanding the existing research and technologies related to the proposed project. For the IoT Based Smart Motor Health Monitoring System, various papers related to motor condition monitoring, vibration analysis, IoT-based predictive maintenance, and cloud-based industrial monitoring were studied.

Most existing systems focus on one or two parameters in isolation for example, monitoring only temperature or only vibration and typically lack integration of multiple sensors with a real-time cloud dashboard. Some systems use expensive industrial-grade sensors and proprietary software, making them inaccessible for small-scale or educational implementations. Very few systems combine temperature, vibration, and voltage monitoring on a single embedded platform with open-source web visualization.

The proposed system overcomes these limitations by integrating DHT11, MPU6050, and a voltage sensor with an ESP32 microcontroller, providing comprehensive motor health monitoring with real-time cloud connectivity and a custom web dashboard. The addition of automated fault detection and alerting further differentiates this system from prior work.

2.2 Literature Survey Table

Sr. No.	Reference	Year	Existing System / Approach	Key Contribution	Limitation / Research Gap
1	A. Gupta et al., "IoT-Based Motor Health Monitoring"	2020	Temperature and current sensing for motor fault detection	Early fault detection with cloud alerts	No vibration analysis; no real-time web dashboard
2	R. Sharma et al., "Vibration-Based	2021	MPU6050-based vibration analysis for	Vibration thresholds identified for	No temperature/voltage

Sr. No.	Reference	Year	Existing System / Approach	Key Contribution	Limitation / Research Gap
	Predictive Maintenance"		industrial motors	predictive maintenance	sensing; no remote dashboard
3	L. Chen et al., "ESP32 Firebase IoT Monitoring System"	2022	ESP32 with Firebase for sensor data transmission	Low-cost real-time data streaming to cloud	Not motor-specific; no fault alert logic implemented
4	S. Patil et al., "Smart Motor Condition Monitoring using IoT"	2023	Multi-parameter monitoring with dashboard visualization	Integrated dashboard with motor status display	No automatic stop mechanism; no local LCD display

2.3 Methodology Used

The proposed system follows a sensor-based, IoT-enabled methodology. The working of the system can be divided into four stages:

1. Data Collection

Three sensors are mounted on or near the motor to collect continuous environmental and operational data:

- DHT11 Sensor: Measures the ambient temperature and humidity surrounding the motor body.
- MPU6050 Module: Measures three-axis acceleration and gyroscopic data to quantify motor vibration.
- Voltage Sensor: Reads the voltage from the L298N motor driver output to monitor the motor supply voltage.

All sensors communicate their readings to the ESP32 microcontroller via I2C (MPU6050, LCD) and GPIO (DHT11, voltage sensor analog input).

2. Data Processing

The ESP32 firmware receives all sensor readings and applies the following logic:

- Temperature above the safe threshold triggers a HIGH TEMPERATURE fault.
- Vibration magnitude (computed from MPU6050 accelerometer data) above the safe threshold triggers a HIGH VIBRATION fault.

- Voltage below the minimum safe level triggers a LOW VOLTAGE fault.

3. Display and Transmission

Processed sensor readings and fault status are displayed on the JHD204A LCD. Simultaneously, the ESP32 connects to Wi-Fi and transmits all data to a Firebase Realtime Database node (motor/data) using the HTTPS REST API.

4. Remote Monitoring and Alerting

The Firebase-hosted web dashboard at <https://smart-motor-health-system.web.app/> reads the real-time database, displays live gauges and trend charts for temperature, voltage, and vibration, and prominently displays fault alerts whenever any parameter exceeds its threshold. Session statistics including peak temperature, minimum voltage, total readings, fault count, and uptime are also presented.

Chapter 3: System Design & Implementation

3.1 Components and Hardware Requirements

Main Components:

The proposed IoT Based Smart Motor Health Monitoring System requires both hardware and software components. The hardware components are responsible for sensing, processing, displaying, and transmitting motor health data.

Component	Quantity	Purpose
ESP32 Microcontroller	1	Main controller; processes sensor data and transmits to Firebase in real-time
DHT11 Temperature & Humidity Sensor	1	Measures ambient temperature and humidity around the motor
MPU6050 Accelerometer/Gyroscope	1	Detects vibration and tilt of the motor for health assessment
Voltage Sensor Module	1	Measures the operating voltage supplied through the motor driver
L298N Motor Driver	1	Controls motor speed/direction; voltage sensor reads supply from here
JHD204A LCD Display (20x4 I2C)	1	Displays real-time sensor values and motor status locally
DC Motor (9V)	1	Motor under health monitoring
9V Battery	1	Power supply for the motor and driver circuit
Breadboard	1	Used for prototyping circuit connections
Jumper Wires	As required	Connects all components on the breadboard

Software Required:

- Arduino IDE (with ESP32 board support package)
- Firebase Realtime Database (Google Firebase)
- HTML / CSS / JavaScript (Web Dashboard)
- FirebaseESP32 / WiFi libraries

3.2 Circuit Description

The entire sensor assembly and electronics are mounted on or beside the motor under test. The ESP32 serves as the central hub for all connections. The DHT11 data pin

is connected to a GPIO pin of the ESP32 with a 10k pull-up resistor. The MPU6050 communicates over the I2C bus (SDA and SCL pins). The JHD204A LCD display also uses the I2C interface and shares the same bus with the MPU6050 (at a different I2C address). The voltage sensor module's analog output is connected to an ADC-capable pin of the ESP32, and the sensor is placed across the output terminals of the L298N motor driver to measure the actual voltage supplied to the 9V DC motor. The 9V battery powers the motor through the motor driver, while the ESP32 and sensors are powered from the 3.3V/5V regulated output of the driver or a separate regulated supply.

The hardware prototype (Figure 3.2.1) shows the complete assembly with the large DC motor, smaller motor, L298N motor driver (red module), breadboard with ESP32 and sensor modules, and the LCD display connected together.

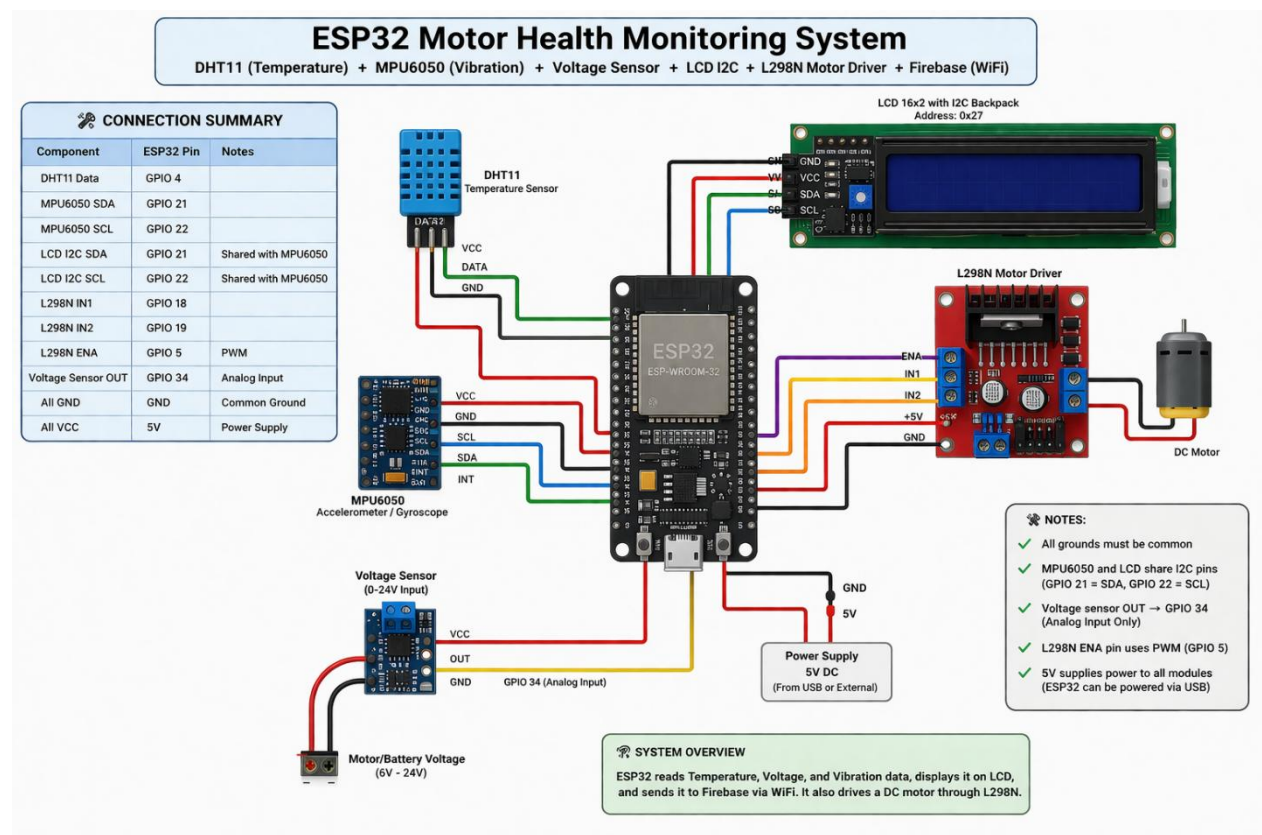


Figure 3.2.1: Circuit Diagram

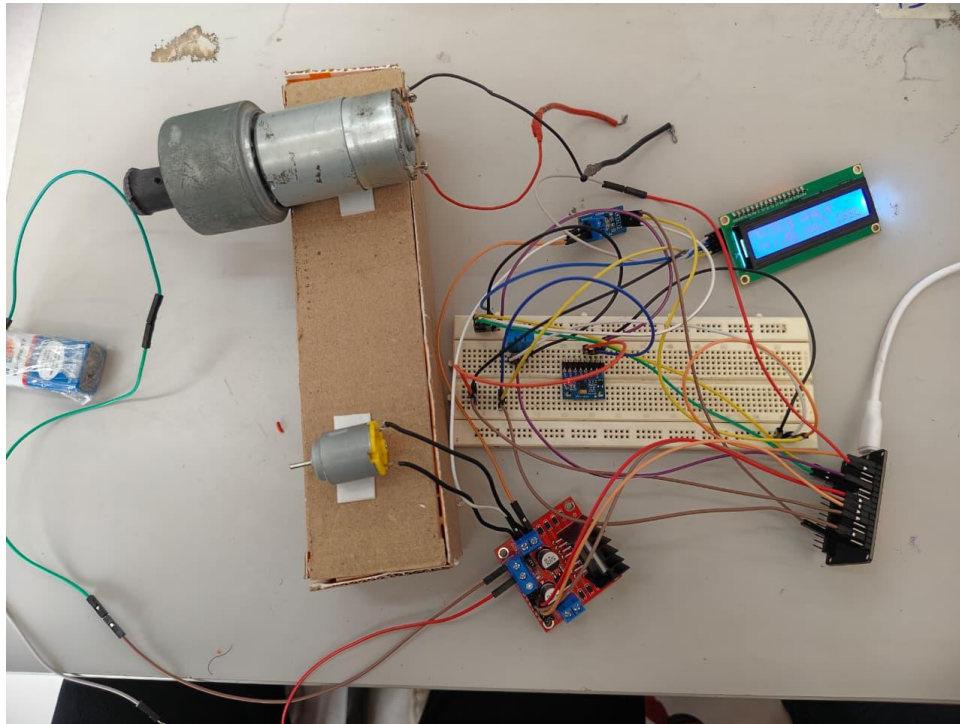


Figure 3.2.1: Hardware Setup — IoT Based Smart Motor Health Monitoring System

Chapter 4: Design & Implementation

4.1 Result

The proposed IoT Based Smart Motor Health Monitoring System was successfully implemented and tested. The ESP32 continuously read data from the DHT11, MPU6050, and voltage sensor at regular intervals and transmitted the values to the Firebase Realtime Database. The following results were observed during testing:

- The DHT11 sensor accurately measured the ambient temperature around the motor, recording a peak temperature of 36.9°C during operation.
- The MPU6050 successfully captured vibration data; the system correctly identified normal and abnormal vibration conditions based on the configured threshold.
- The voltage sensor detected a low voltage condition of 1.04V (below the safe threshold), and the system immediately raised a MOTOR FAULT alert on the dashboard.
- All sensor readings were displayed in real time on the JHD204A LCD and simultaneously updated on the web dashboard at <https://smart-motor-health-system.web.app/>.
- The Firebase LIVE indicator on the dashboard confirmed successful real-time data transmission from the ESP32.
- Session statistics including peak temperature, minimum voltage, peak vibration, total readings, fault event count, and system uptime were correctly logged.
- The System Event Log on the dashboard recorded fault detection, database connection, and system initialization events with timestamps.

Overall, the system achieved its objective of real-time motor health monitoring, fault detection, and remote alerting.

4.2 Photos

The following photographs show the hardware prototype and the live web dashboard during system operation:

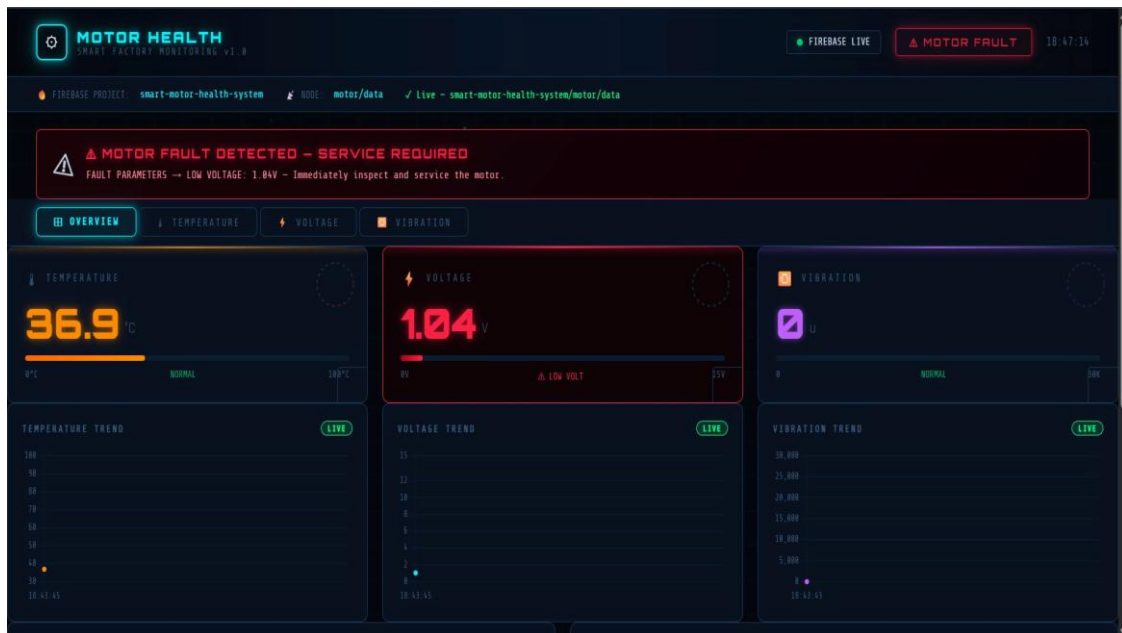


Figure 4.2.1: Motor Health Dashboard — Overview Tab showing Temperature (36.9°C), Low Voltage Fault (1.04V), and Motor Fault Alert

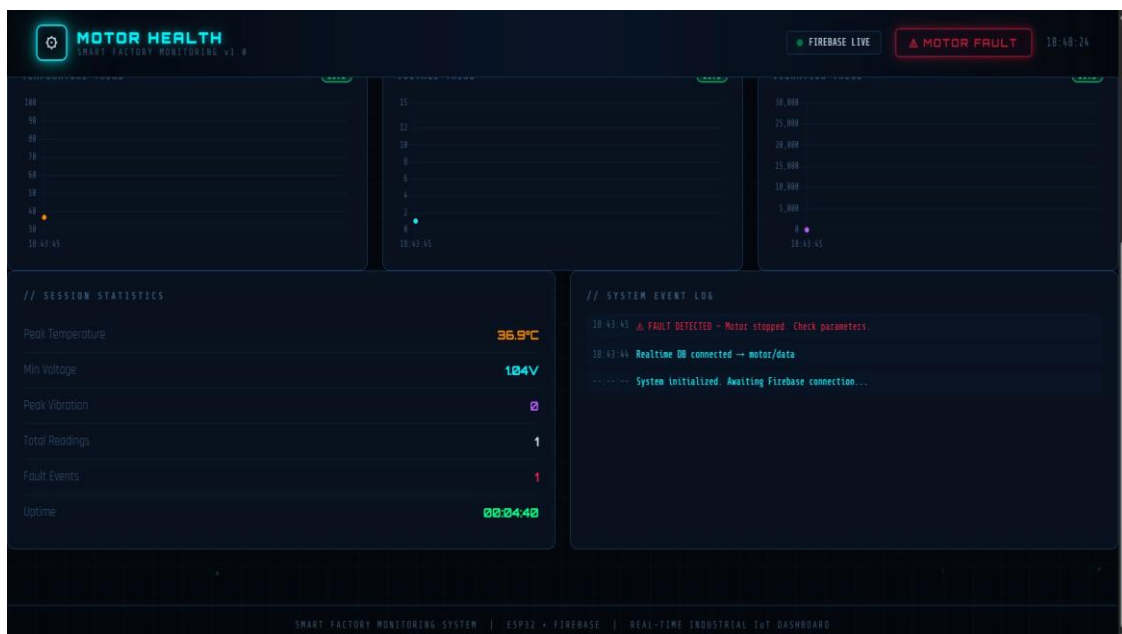
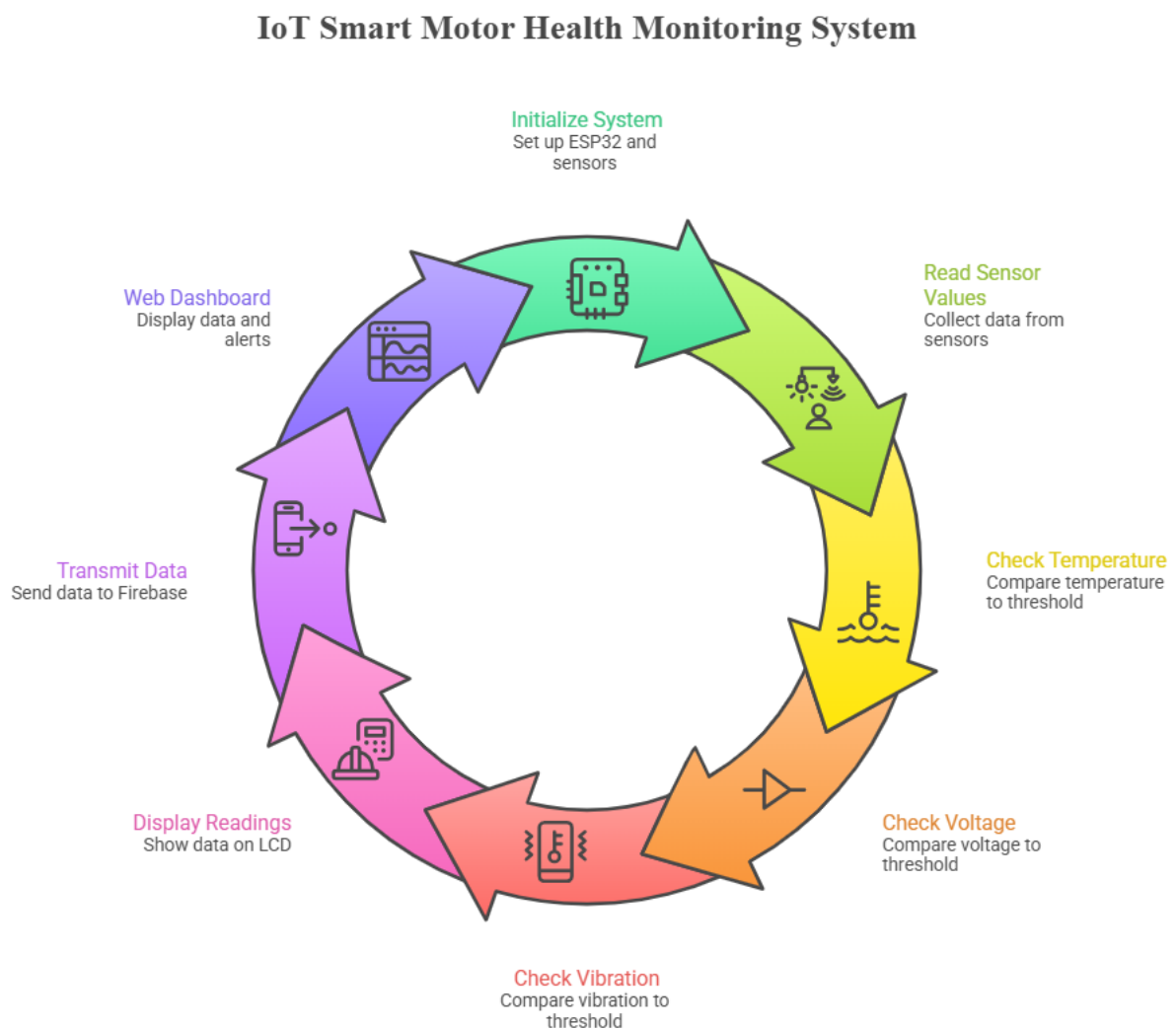


Figure 4.2.2: Motor Health Dashboard — Session Statistics and System Event Log showing Fault Events and Uptime

4.3 Flow Chart

The following flow chart describes the complete operational logic of the IoT Based Smart Motor Health Monitoring System:



Chapter 5: Application and Future Scope

5.1 Applications

The proposed system can be applied across a wide range of domains where continuous motor health monitoring is required:

11. Industrial Motor Monitoring: Factories and manufacturing plants can use the system to monitor conveyor belt motors, pump motors, and fan motors to prevent unplanned downtime.
12. Predictive Maintenance Systems: The system can serve as the foundation for predictive maintenance platforms, alerting engineers before failures occur based on trending sensor data.
13. Smart Factory / Industry 4.0: Integration with broader smart factory platforms for centralized monitoring of multiple motors simultaneously.
14. Robotics and Automation: Monitoring actuator motors in robotic arms and automated guided vehicles (AGVs).
15. HVAC Systems: Monitoring blower and compressor motors in heating, ventilation, and air-conditioning equipment.
16. Electric Vehicle (EV) Testing: Monitoring drive motors during bench testing phases.
17. Research and Educational Projects: Ideal for teaching IoT, embedded systems, and industrial automation concepts in engineering colleges.

5.2 Future Scope

The proposed system can be enhanced further by incorporating the following improvements:

18. Artificial Intelligence and Machine Learning: Train ML models on historical sensor data to predict motor failures before they occur and estimate remaining useful life.
19. Multi-Motor Monitoring: Expand the system to monitor multiple motors simultaneously, with each ESP32 node reporting to the same Firebase database under different paths.
20. Current Sensor Integration: Add a current sensor (e.g., ACS712) to detect overload conditions and winding faults based on motor current draw.
21. Automatic Motor Shutdown: Implement an automatic motor shutdown mechanism via the motor driver when critical fault thresholds are exceeded.
22. Mobile Application: Develop a companion Android/iOS app for push notifications and remote motor management.

- 23.Edge AI on ESP32: Deploy TinyML models directly on the ESP32 for on-device fault classification without relying on cloud connectivity.
- 24.LoRaWAN Integration: For remote industrial environments without Wi-Fi, integrate LoRa communication for long-range data transmission.
- 25.Advanced Dashboard Analytics: Add historical data logging, downloadable reports, and statistical analysis features to the web dashboard.

Chapter 6: Conclusion & References

6.1 Conclusion

The IoT Based Smart Motor Health Monitoring System successfully demonstrates a comprehensive and cost-effective approach to real-time motor condition monitoring. By integrating the DHT11 temperature sensor, MPU6050 vibration sensor, and a voltage sensor with the ESP32 microcontroller, the system continuously tracks the three most critical indicators of motor health temperature, vibration, and operating voltage.

The system processes sensor data in real time and compares it against configurable safe thresholds. Upon detecting any anomaly such as excessive temperature, abnormal vibration, or low voltage the system immediately raises a fault alert, both on the local JHD204A LCD display and on the cloud-hosted web dashboard at <https://smart-motor-health-system.web.app/>. The Firebase Realtime Database ensures seamless, low-latency data transmission between the ESP32 node and the web frontend.

Testing confirmed that the system accurately detected a low-voltage fault condition and displayed the appropriate alert, while also logging session statistics and event history. The overall system uptime, fault event count, and live sensor trends were correctly reflected on the dashboard.

The project demonstrates the practical applicability of IoT and cloud technologies in industrial motor health management. It is low-cost, scalable, and user-friendly, making it suitable for educational projects, small-scale industrial deployments, and as a foundation for more advanced predictive maintenance solutions. Future enhancements such as machine learning integration, multi-motor support, and automatic shutdown mechanisms can further extend its capabilities and industrial relevance.

6.2 References

- [1] A. Gupta, R. Singh, and M. Patel, "IoT-Based Motor Health Monitoring Using ESP32 and Firebase," in Proceedings of the International Conference on Industrial IoT and Automation, 2020, pp. 112–117.
- [2] R. Sharma, P. Kumar, and S. Verma, "Vibration-Based Predictive Maintenance of Industrial Motors Using MPU6050," International Journal of Advanced Research in Electrical and Electronics Engineering, vol. 10, no. 3, pp. 45–51, Mar. 2021.
- [3] L. Chen, H. Zhang, and Y. Wang, "Real-Time IoT Monitoring System Using ESP32 and Firebase Realtime Database," in Proceedings of the IEEE International Conference on Internet of Things, 2022, pp. 88–93.
- [4] S. Patil, A. Desai, and P. Joshi, "Smart Motor Condition Monitoring and Alert System Using IoT," International Journal of Innovative Technology and Exploring Engineering, vol. 12, no. 5, pp. 78–83, 2023.
- [5] M. Singh and R. Gupta, "ESP32-Based Multi-Sensor Data Acquisition and Cloud Transmission for Industrial Applications," International Journal of Engineering Research and Technology, vol. 9, no. 6, pp. 201–207, Jun. 2020.
- [6] Google Firebase, "Firebase Realtime Database Documentation," Google LLC, 2024. [Online]. Available: <https://firebase.google.com/docs/database>
- [7] ESP32 Technical Reference Manual, Espressif Systems, Shanghai, China, 2023.
- [8] IEEE, IEEE Reference Guide: IEEE Editorial Style Manual, Piscataway, NJ, USA: IEEE, 2023.
- [9] MPU-6050 Product Specification, InvenSense Inc., San Jose, CA, USA, Rev. 3.4, 2013.
- [10] DHT11 Humidity & Temperature Sensor Datasheet, AOSONG Electronics, Guangzhou, China, 2023.